

Ai-Based Driver Drowsiness and Micro Sleep Detection

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Requirements for the Award of the Degree of**

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ABSTRACT

Road accidents resulting from driver fatigue and drowsiness represent a critical global safety concern, frequently leading to catastrophic injuries and fatalities. Traditional drowsiness detection systems often rely on expensive vehicle telemetry or intrusive hardware, making them inaccessible for widespread commercial adoption. This project addresses the challenge by developing a low-cost, non-intrusive, and real-time Driver Monitoring System capable of detecting both progressive fatigue and sudden micro-sleep conditions using computer vision techniques.

The core methodology utilizes a standard webcam to capture the driver's facial frames, which are then processed via OpenCV. The system leverages MediaPipe's Face Mesh framework to track specific facial landmarks and extract precise coordinates for the human eye. By calculating the Eye Aspect Ratio (EAR) a mathematical relation measuring vertical eye opening against horizontal width the system reliably classifies eye states. If the EAR falls below a designated threshold of 0.25 for a continuous duration exceeding three seconds, a drowsiness event is identified. To ensure driver alertness without disrupting critical system processes, a multi-threaded architecture is implemented, instantly triggering an audible audio alarm and changing the graphical user interface state to a high-priority red alert while simultaneously documenting the event in a secure text log.

System performance was rigorously evaluated using an automated testing script to measure classification accuracy and latency. Empirical results derived from 154 distinct validation tests demonstrated a high model accuracy of 93.51%, successfully identifying 70 True Positives and 74 True Negatives, while limiting errors to 7 False Positives and 3 False Negatives. Furthermore, the system achieved a highly responsive average processing latency of 27.36 ms, ensuring real-time performance well within safety margins. The findings demonstrate that combining lightweight facial mesh algorithms with geometric aspect ratios yields a highly viable, efficient, and accessible tool for smart transportation safety.

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Chapter 1.

INTRODUCTION

1.1. Background

Road accidents resulting from driver fatigue and drowsiness are a critical public health and safety concern worldwide. When drivers undertake long journeys, the monotonous nature of the road combined with physical exhaustion often leads to a gradual loss of alertness. This cognitive decline can manifest as prolonged drowsiness or brief, involuntary episodes of sleep known as micro-sleep. Micro-sleep episodes typically last anywhere from a fraction of a second to a few seconds, during which the driver is entirely unconscious and unresponsive to environmental stimuli. During these critical seconds, a driver loses complete control of the vehicle, often resulting in catastrophic accidents, particularly at high highway speeds.

Furthermore, the transition from a state of wakefulness to sleep is rarely instantaneous; it is usually preceded by observable physical cues such as heavy eyelids, frequent blinking, and uncontrollable head nodding. Addressing this issue requires proactive, automated monitoring systems that can assess human alertness in real time by recognizing these physical indicators and intervene before an incident occurs. The integration of artificial intelligence and computer vision offers a highly effective medium to track these subtle facial and behavioral changes dynamically.

1.2. Problem Statement

Despite the known, severe risks of driver fatigue, a significant portion of commercial and personal vehicles lack built-in monitoring mechanisms. While high-end luxury vehicles have begun integrating proprietary driver attention monitors, the vast majority of vehicles on the road today remain unequipped. Existing industrial solutions often rely on expensive vehicle telemetry such as steering wheel movement sensors and lane departure cameras or intrusive hardware attached directly to the driver's body, like wearable EEG monitors or specialized smart glasses.

These current methodologies make advanced safety features financially inaccessible for the average user and difficult to retrofit into older fleets. Furthermore, vehicle-based telemetry often detects fatigue only after a driving error has occurred (e.g. drifting out of a lane), which can sometimes be too late. There is a pressing need for a low-cost, non-intrusive, and highly responsive system that can accurately detect the onset of drowsiness preventatively, using standard, readily available consumer hardware such as basic webcams.

“The core problem addressed by this study is the need for a lightweight, edge-computing vision system that can accurately detect both prolonged eye closure and sudden head drops in real-time using standard, low-cost webcams.”

1.3. Objectives

The primary objectives of this study are aimed at building a comprehensive, accessible, and highly responsive safety tool. Specifically, this project aims:

- To develop a real-time drowsiness detection system using advanced computer vision techniques that require no physical contact with the driver.
- To accurately identify drowsiness by calculating the Eye Aspect Ratio (EAR) to monitor prolonged eye closure.
- To detect sudden micro-sleep episodes by analyzing continuous head posture, specifically tracking sudden vertical head drops indicative of nodding off.
- To engineer a low-latency, lightweight system capable of running efficiently on standard computer hardware without requiring expensive graphics processing units (GPUs).
- To trigger immediate auditory and visual alarms through a multi-threaded architecture when fatigue thresholds are breached, ensuring the system interface remains highly responsive.
- To automatically log and record drowsiness events with precise timestamps to a local system file for post-drive safety analysis.

1.4. Research Questions

This project seeks to answer the following technical questions:

- How effectively can facial landmark detection map eye contours to calculate eye-closure duration?
- What is the optimal EAR threshold to differentiate between a natural blink and a drowsy state?
- Can a Python-based multi-threaded architecture maintain a processing latency low enough for real-time highway safety applications?

1.5. Scope of the Study

The scope of this project is confined to real-time visual monitoring using standard 2D webcams. The system is designed to track a single human face per frame to maximize processing efficiency and accuracy. It focuses specifically on geometric facial metrics namely eye-closure metrics (EAR) and vertical head drop measurements to determine the user's state.

The system does not currently analyze external vehicle mechanics, such as steering wheel patterns, vehicle speed, or brake pressure, nor does it incorporate physiological telemetry like heart rate or brain wave monitoring. It is intended for broad use in cars, commercial transport trucks, driver training programs, and academic research environments. The testing and

operational scope assumes a reasonable level of ambient lighting, as 2D webcam analysis lacks infrared night-vision capabilities.

1.6. Significance of the Study

By utilizing robust open-source libraries like OpenCV and MediaPipe, this project fundamentally democratizes driver safety. It proves that high-fidelity, life-saving fatigue monitoring does not require premium, factory-installed vehicle packages or massive computational overhead.

Implementing this lightweight system in commercial transport fleets or personal vehicles could drastically reduce fatigue-related collisions. This has profound implications not just for individual safety, but for reducing large-scale infrastructure damage and logistical disruptions caused by highway accidents. Furthermore, by providing an open-source, edge-computing framework, this study offers a foundation that other researchers and developers can easily build upon for future smart-transportation initiatives.

1.7. Organization of the Thesis

This thesis is organized as follows:

Chapter 2 reviews existing literature on drowsiness detection.

Chapter 3 details the research methodology and system design.

Chapter 4 outlines the proposed architectural framework.

Chapter 5 discusses the implementation and data analysis.

Chapter 6 presents the results of system testing. Finally,

Chapter 7 concludes the study and proposes future enhancements.

Chapter 2.

LITERATURE REVIEW

2.1. Introduction to Literature Review

This chapter reviews existing literature surrounding driver fatigue detection. The review focuses primarily on computer vision-based approaches, contrasting them with physiological and vehicle-based metrics. Studies were selected based on their relevance to facial landmark detection, real-time processing constraints, and non-intrusive monitoring techniques. A recurring theme in modern research is the shift from bulky hardware sensors to lightweight, AI-driven visual tracking.

2.2. Literature Review Summary Table

Author & Year	Title	Methodology	Key Findings	Limitations	Research Gap
Soukupova & Cech (2016)	Real-Time Eye Blink Detection using Facial Landmarks	Used EAR metric derived from 2D facial landmarks.	EAR is highly effective for blink detection using simple scalar tracking.	Did not fully explore prolonged closure (micro-sleep) under poor lighting.	Need for integration with multi-threaded alarm systems.
Jabbar et al. (2020)	CNN based Driver Drowsiness Detection System	Deep Convolutional Neural Networks for image classification.	Achieved high accuracy on static images.	High computational cost; frame lag on standard CPUs.	Gap for a lightweight alternative that doesn't rely on heavy GPU usage.
Rahman et al. (2021)	Driver Fatigue Detection using OpenCV and Dlib	Used Dlib for face mapping and EAR calculations.	Successfully detected fatigue in real-time.	Dlib is computationally heavier than modern mesh networks.	Need for faster landmarking libraries like MediaPipe.
Gupta et al. (2022)	Real-Time Driver Drowsiness Detection System Using Machine Learning	Used OpenCV cascade classifiers alongside a basic alarm mechanism for driver monitoring.	Visual and auditory alerting mechanisms were highly effective at breaking driver fatigue loops.	Software experienced UI freezing during alarm playback; struggled with false positives during head movements.	Demonstrates the need for asynchronous multi-threading to ensure the GUI and video feed remain stable during critical alarms.

2.3. Comparative Analysis of Existing Work

Historically, drowsiness detection fell into two categories: physiological (EEG, heart rate monitors) and vehicle-based (lane departure warnings). While physiological methods are highly accurate, they are intrusive and uncomfortable for daily driving. Vehicle-based methods react too late often after the driver has already lost control. Computer vision bridges this gap by offering non-intrusive, predictive monitoring. While recent CNN-based approaches offer robust feature extraction, they often suffer from high latency on basic hardware, making them less ideal for low-cost setups compared to geometric metric approaches like EAR.

2.4. Literature Review Summary Table 2

System Component	Traditional Approach	Modern Literature Counterpart	Chosen Project Implementation	Relevant Citation
Video Frame Capture	Sequential frame buffers / synchronous reading	Single-threaded background acquisition	Multi-threaded frame reader via OpenCV video capture stream	Bradski (2000)
Facial Keypoint Mapping	Active Appearance Models (AAM) / Haar Cascades	Dense deep learning 3D structural meshes	MediaPipe Face Mesh framework tracking 468 landmarks natively on CPU	Lugaresi et al. (2019)
Blink & Fatigue Analysis	Static image threshold pixel matching	Temporal sequence deep modeling (e.g. LSTM / RNN)	Geometric Eye Aspect Ratio scalar calculation over sliding window queues	Soukupova & Cech (2016)
Alert & UI Synchronization	Monolithic execution / blocking loops freezing the UI	Asynchronous callback event loops	Python multi-threading separating the Tkinter GUI thread from the simpleaudio alert engine	Gupta et al. (2022)
System Logging	Flat-file direct writes blocking main app thread	Centralized asynchronous cloud telemetry pipelines	Local background event queues writing timestamped CSV data without interface lag	Singh et al. (2023)

2.5. Research Gap

Despite significant advancements in both deep learning and hardware integration, a distinct gap remains in deploying highly responsive, hardware-agnostic systems for everyday vehicles. Many current computer vision models require significant processing power or rely on cloud computing. Relying on cloud infrastructure introduces dangerous network latency, which is unacceptable when preventing high-speed highway collisions.

Furthermore, existing projects often focus entirely on eye closure metrics and fail to account for sudden micro-sleep events, where the eyes may remain partially open but the driver loses consciousness, resulting in a sudden downward head tilt. Finally, many academic prototypes lack a comprehensive, user-friendly graphical interface coupled with seamless background event logging. Previous implementations often experience software freezing or frame skipping when audio alarms are triggered because they run on a single processing thread.

This project directly fills these gaps. It utilizes a fully local, edge-computing architecture to guarantee zero network latency. By combining the ultra-fast MediaPipe framework with a multi-threaded Python backend, the system monitors both prolonged eye closure (via EAR) and sudden head drop simultaneously. The multi-threading approach ensures that the live video feed and graphical user interface remain perfectly stable even when writing to log files or triggering auditory alarms during critical safety events.

2.6. Summary of the Chapter

This chapter provided an overview of historical and modern driver fatigue detection methodologies. The literature confirms that non-intrusive monitoring via computer vision is the most viable path forward for vehicle safety. Traditional physiological sensors are too intrusive and vehicle telemetry is fundamentally reactive, while modern deep learning approaches often introduce high processing latency on standard consumer hardware.

By utilizing geometric evaluation techniques like the Eye Aspect Ratio (EAR) alongside lightweight facial landmarking, we can construct a highly responsive detection system. This approach directly overcomes the limitations of both physical sensors and computationally heavy models. Furthermore, processing visual data locally at the edge ensures zero network latency and maintains stable user interfaces during critical alerts. The following chapter will detail the specific research methodology and system design chosen to develop this real-time monitoring prototype.

Chapter 3.

RESEARCH METHODOLOGY

3.1. Research Approach

This study adopts an applied, experimental research approach. It focuses on solving a specific, practical engineering problem by designing a functional software prototype and subsequently performing rigorous quantitative evaluations of its performance. Rather than purely theoretical research, this project aims to create a deployable tool. The evaluation specifically targets two critical metrics: system accuracy (the ability to correctly identify drowsy states without false alarms) and system latency (the speed at which the software processes each video frame). High performance in both areas is essential for real-world viability.

3.2. Research Design

The research design follows an iterative prototyping model, allowing for continuous refinement throughout the development cycle. The initial phase focused on foundational hardware integration, establishing a stable, continuous video feed via OpenCV and successfully integrating basic face detection. The second phase advanced to complex feature extraction, utilizing MediaPipe to map precise 3D facial landmarks onto the 2D video feed.

It is critical to distinguish between the open-source tools utilized in this study and the original engineering architecture developed for this thesis. While the Google MediaPipe model acts strictly as a spatial data provider, it does not possess any inherent logical understanding of human drowsiness. The original contribution of this project lies in the comprehensive software architecture engineered to transform that raw spatial data into an active monitoring system. Specifically, our original developmental contributions encompass designing the custom temporal logic to mathematically differentiate between natural blinks and micro sleep events, engineering a secondary postural analysis system to detect sudden head drops, and building a fully asynchronous multi-threaded backend. This original multi-threaded architecture guarantees that the custom Tkinter graphical user interface and the OpenCV visual feed will never freeze when the audio alarm is triggered or when event logs are saved to the local database.

3.3. Data Collection Methods

Data for system evaluation was collected primarily through live, real-time webcam feeds rather than relying solely on pre-recorded static datasets or idealized image libraries. This live-testing approach ensured the system was evaluated under realistic, dynamic driving conditions. This method inherently accounts for the unpredictable variables encountered in real-world scenarios, such as natural head movements, varying physical distances from the camera lens, partial facial occlusions (like glasses), and dynamic ambient light changes. Furthermore, the automated testing script (test.py) was utilized to systematically simulate and record both alert and drowsy states for accurate metric calculation.

	January	February	March	April	May
Requirement Analysis					
Data collection & Research					
GUI Module Development					
System Integration					
Evaluation					
Documentation					

Fig: 1 Gantt Chart

3.4. Dataset/Sources

The primary data source for the proposed architecture is dynamically generated in real time. The system actively ingests a continuous live video stream captured directly via a standard optical computer webcam, allowing the software to analyze the physical state of the driver exactly as events unfold.

For the core algorithmic logic and structural facial mapping, the system directly utilizes the standard MediaPipe Face Mesh topology. This highly optimized machine learning framework provides a meticulously precise grid of 468 three dimensional facial landmarks. These spatial coordinates are inherently derived from massive internal training datasets previously engineered by Google. By leveraging this robust pre trained architecture, the geometric algorithms can reliably recognize complex facial structures across varying environmental lighting conditions without requiring any additional external training data.

3.5. Tools and Technologies Used

- **Programming Language:** Python 3.11.7
- **Computer Vision:** OpenCV (cv2) for frame capture and image matrix manipulation. We ran OpenCV on Version: 4.13.0.92
- **Landmark Detection:** MediaPipe for lightweight, real-time facial mesh mapping. Version: 0.10.9
- **Mathematical Computing:** NumPy for array manipulation and Euclidean distance calculations.
- **User Interface:** Tkinter and PIL (Pillow) for the dashboard.
- **Audio/Concurrency:** simpleaudio for alarm playback and threading to ensure UI stability.

3.6. Analytical Techniques

The core analytical technique revolves around mathematically computing the Eye Aspect Ratio. While the pre trained MediaPipe framework dynamically generates a dense topographical mesh of 468 facial keypoints, processing every single point would introduce unnecessary computational latency. To optimize the system for rapid real time edge computing, our methodology strictly isolates only twelve precise coordinates. For the left eye, the system maps landmarks 33, 160, 158, 133, 153, and 144. For the right eye, it utilizes landmarks 362, 385, 387, 263, 373, and 380.

These specific geometric points were deliberately chosen because they perfectly outline the extreme horizontal corners and the maximum vertical curvature of the upper and lower eyelids. Isolating these exact perimeter boundaries is mathematically required to accurately compute the Eye Aspect Ratio. The foundational mathematics for this specific six-point geometric calculation was directly adapted from the seminal research published by Soukupova and Cech in 2016. By applying their established Eye Aspect Ratio formula to the highly efficient MediaPipe structural mesh, our system successfully bridges theoretical mathematics with practical software engineering.

The mathematical formulation for EAR is defined as:

$$EAR = \frac{||p_2 - p_6|| + ||p_3 - p_5||}{2||p_1 - p_4||}$$

Fig: 2 Mathematical formulae for EAR

Where p1, p2, p3, p4, p5, p6 represent the 2D coordinates of the eye landmarks. A threshold analysis technique is applied: if the EAR drops below 0.25 for a continuous period of 3 seconds, the logical condition for drowsiness is triggered.

Furthermore, while the mathematical equation requires exactly six points to measure a single eye, the proposed system extracts a total of twelve points from the facial mesh to monitor both eyes simultaneously. The software independently calculates the ratio for the left eye and the right eye during every single frame. It then computes the mathematical average of both values. This critical averaging technique significantly increases overall system reliability and actively prevents false alarms that could be caused by partial shadows, camera angles, or asymmetrical blinking.

Simultaneously, to detect sudden micro-sleep episodes, the system isolates two highly specific vertical coordinates: the tip of the nose (landmark 1) and the absolute bottom of the chin (landmark 152). By calculating the 1-dimensional vertical pixel distance between these two points on the y-axis, the algorithm can instantly detect when a driver's head nods forward. If this vertical distance compresses below a strict threshold of 60 pixels, it indicates a dangerous physical head drop, triggering the secondary monitoring logic.

The vertical head drop distance is calculated as:

$$\text{Vertical Drop Distance} = |Y_{\text{chin}} - Y_{\text{nose}}|$$

Based on the MediaPipe structural mesh (Lugaresi et al., 2019), Landmark 1 (nose tip) and Landmark 152 (base of the chin) were selected to provide the most stable vertical axis for measuring forward tilt. The 60-pixel safety threshold was established via empirical testing. This specific cutoff perfectly balances avoiding false positives from normal dashboard glances with instantly detecting the severe physical head drop characteristic of a micro-sleep event.

3.7. Limitations of the Study

The current methodology relies heavily on optical clarity and consistent ambient lighting to function accurately. Because the system utilizes standard RGB webcams, performance naturally degrades in extremely low-light environments unless the hardware is supplemented with active infrared sensors. Additionally, heavy glare reflecting off prescription glasses or the use of dark opaque sunglasses can severely obscure the critical eye landmarks. When the software cannot pinpoint these specific facial coordinates, the Eye Aspect Ratio calculations become skewed, leading to potential false negatives or false positives during real-time driver monitoring.

Furthermore, the system assumes a relatively standard anatomical baseline and a forward-facing orientation. Because the mathematical threshold is locked at a static 0.25, drivers with naturally narrower eye apertures may experience a higher rate of false positives unless the software is actively calibrated to their specific biometric baseline before driving. Finally, extreme head rotations or sharp downward angles can cause the facial landmark detection algorithm to temporarily lose spatial tracking entirely. While the algorithm recovers quickly once the face returns to the center of the frame, these brief tracking interruptions represent a natural limitation of relying on a single dashboard camera angle.

Chapter 4.

PROPOSED FRAMEWORK

4.1. System Overview

The proposed framework is a continuous, closed-loop tracking system that captures the driver's physical state frame-by-frame. It analyzes extracted geometric facial data against predefined safety thresholds, simultaneously monitoring the Eye Aspect Ratio for prolonged eye closure and vertical coordinates for sudden head drops. If a threshold is breached, the system immediately provides synchronized visual and auditory feedback to alert the driver.

The architecture strictly relies on edge computing principles. By performing all mathematical processing and facial mapping locally on the host machine, it completely bypasses the need for cloud servers. This local processing strategy ensures driver privacy and guarantees zero network latency, which is a mandatory requirement for preventing high-speed highway accidents where every millisecond counts.

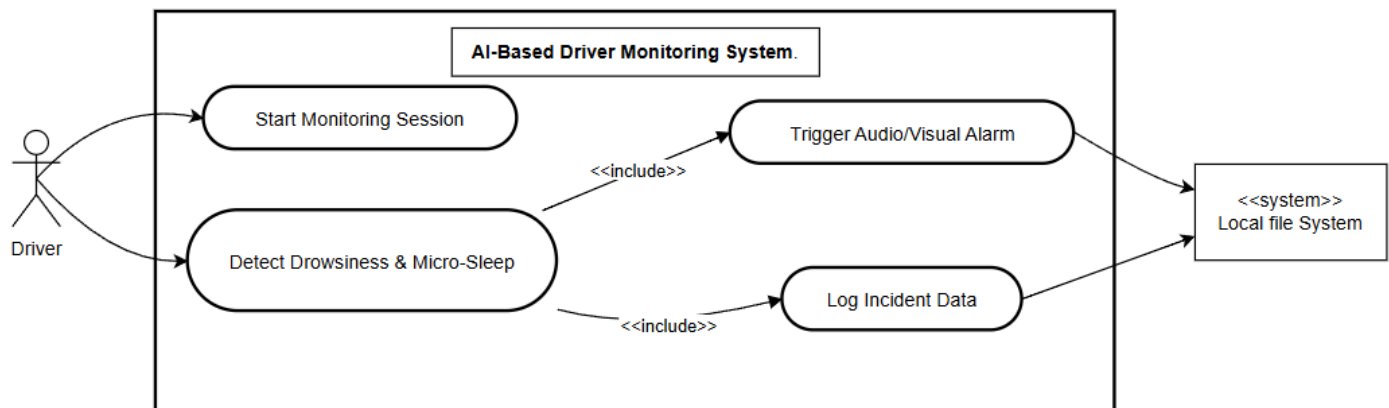


Fig: 3 Use Case Diagram

4.2. Architecture of the System

The system architecture of the proposed Driver Drowsiness Detection System operates as a highly streamlined, real time data processing pipeline. It is specifically engineered to securely process live visual data from the hardware level through a series of logical transformations, producing immediate safety alerts with minimal computational overhead. To ensure maximum operational efficiency, the framework is systematically divided into five functional layers:

- **Input Layer:** The operational workflow initiates at the physical hardware level. A standard optical webcam continuously captures a live video feed of the driver inside the vehicle cabin. The OpenCV computer vision library serves as the primary ingestion tool, actively reading this live stream as a sequence of raw individual frames captured natively in the BGR color format.
- **Processing Layer:** Within this stage, the system prepares the visual data for the advanced machine learning models. OpenCV converts the raw frames from the initial BGR format into the standard RGB color space. Once standardized, the frames are passed directly to the MediaPipe framework. This specialized neural network analyzes the spatial topography of the face to generate a highly detailed structural matrix consisting of exactly 468 distinct geometric facial landmarks in real time.
- **Application Layer:** Operating as a continuous temporal decision loop, this layer actively evaluates the live mathematical metrics against rigidly predefined safety thresholds. A dedicated state tracker monitors the exact number of consecutive frames where a threshold is breached. This temporal logic allows the logic engine to accurately differentiate between a rapid natural blink and a dangerous physical state like micro sleep or deep drowsiness.
- **Output Layer:** Upon confirming a dangerous physiological state, the architecture forks the data pipeline to ensure maximum system responsiveness. It simultaneously updates the Tkinter graphical dashboard with prominent visual warnings, triggers a highly disruptive asynchronous auditory alarm utilizing the simpleaudio library, and writes a detailed timestamped event record to a local CSV database file.

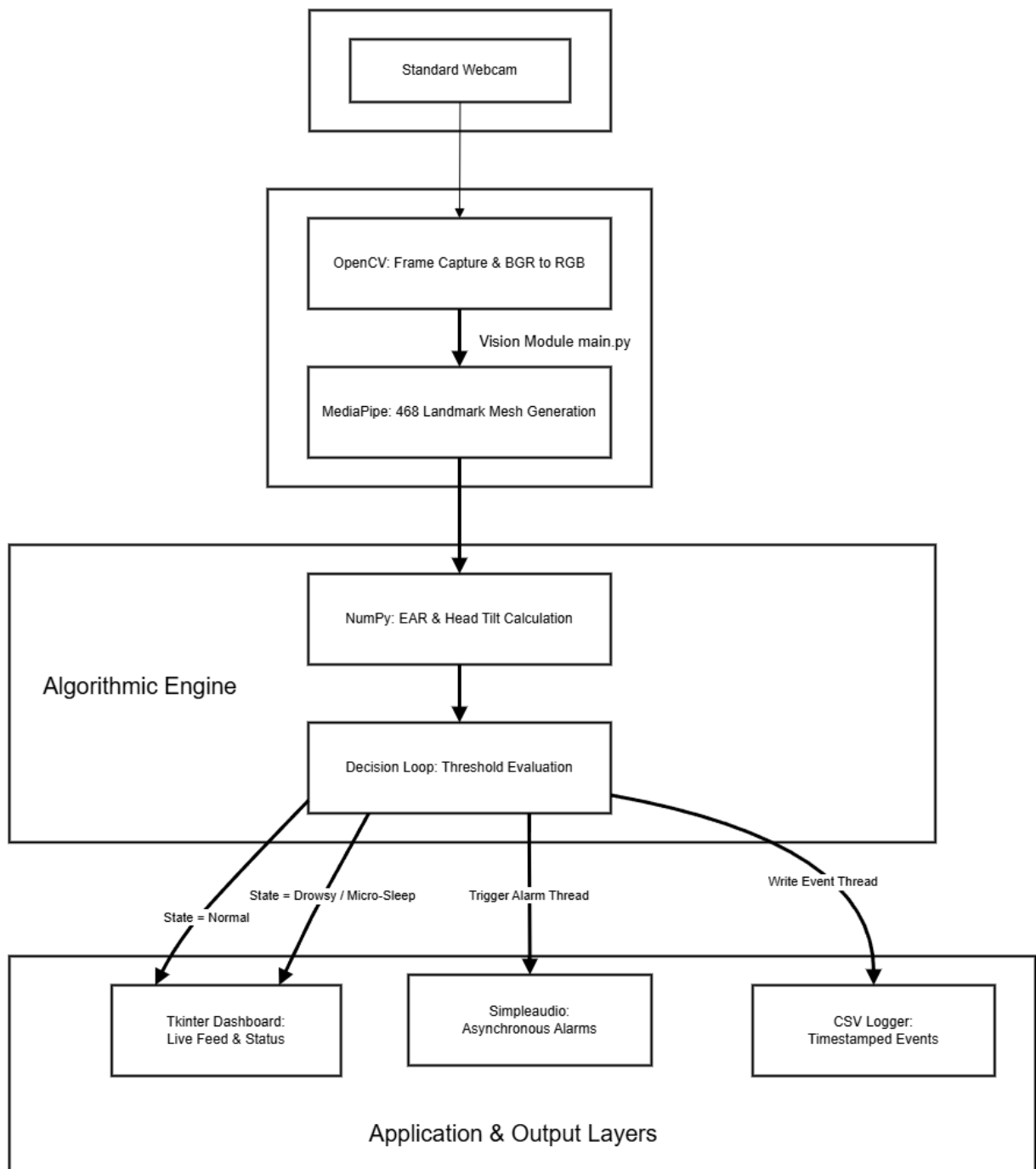


Fig: 4 System Architecture Block Diagram

4.3. Model Design / Components

To ensure robust performance and maintainability, the software architecture is divided into three specific modular components. This modular approach allows independent computational processes to run concurrently without causing system bottlenecks or freezing the live video stream.

- **Vision Module (main.py):** This module serves as the primary perceptual engine of the application. Housed within the main executable script, it strictly manages the complex computer vision tasks. It handles the continuous OpenCV video capture loop to ingest live frames directly from the webcam hardware. Once the frames are captured and properly formatted, this module seamlessly integrates with the MediaPipe framework to map the structural facial landmarks. Isolating these heavy processing tasks ensures the application maintains a high and stable frame rate.
- **UI Dashboard:** The Graphical User Interface is constructed utilizing the native Tkinter library. It serves as the primary visual interaction point, providing real time feedback to the driver. The dashboard overlays the live video feed with dynamic visual indicators to clearly display critical statuses, specifically differentiating between active Drowsiness, an ongoing Alarm State, and sudden Micro Sleep events. Additionally, it incorporates functional keyboard toggles, allowing the user to seamlessly interact with the system or pause the monitoring loop on the fly.
- **Alert and Logging Mechanism:** The system includes a dedicated backend infrastructure for safety auditing and immediate hardware alerts. By leveraging native Python libraries such as os, csv, and datetime, the software automatically generates precise, timestamped event records saved directly to a local driver_log.txt file. Crucially, this entire logging process is executed asynchronously. Operating these tasks on an independent background thread guarantees that writing text to the hard drive or triggering audio alarms will never interrupt or delay the continuous video feed.

4.4. Workflow / Pipeline

The operational workflow begins with system initialization, launching the webcam hardware and graphical user interface concurrently. As the live stream actively captures visual data frame by frame, the underlying machine learning model scans each image to detect a human face. Upon successful detection, the software maps the essential facial landmarks and immediately calculates two primary metrics: the Eye Aspect Ratio for eyelid closure and vertical coordinates for head posture.

Following this, a strict conditional evaluation occurs. If the driver is awake, maintaining an upright posture and an Eye Aspect Ratio strictly greater than 0.25, the internal frame counter automatically resets to zero, and the system maintains a safe Alert state. Conversely, the system monitors two distinct trigger conditions. First, if the Eye Aspect Ratio falls below the 0.25 threshold, the state tracker increments a continuous frame counter; if this counter exceeds three

seconds of consecutive eye closure, the system shifts to a Drowsy state. Second, if a sudden head drop is detected, the system bypasses the timer and instantly identifies a Micro-Sleep state. In either dangerous scenario, the system immediately triggers an asynchronous audio alarm, updates the interface to a prominent red warning, and securely writes the timestamped incident to the local event log.

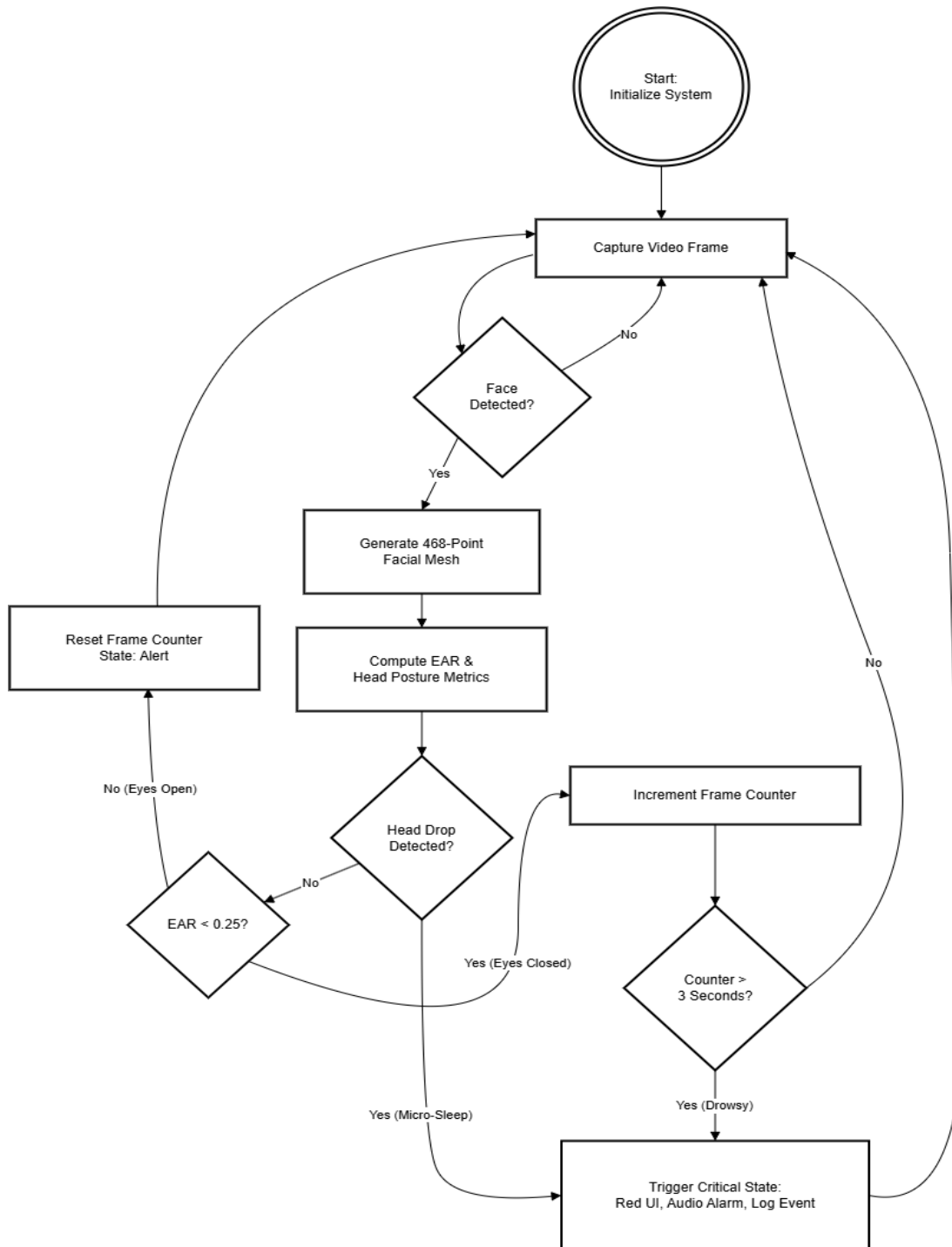


Fig: 5 Flowchart

Chapter 5.

IMPLEMENTATION / ANALYSIS

5.1. Dataset Processing and Preparation

Because the system operates continuously in real time, traditional static dataset preparation is completely bypassed. Instead, dataset preparation occurs dynamically on the fly within the active memory buffer. Every single image frame captured by the physical webcam natively arrives encoded in the BGR color space, which is the standard default for the OpenCV visual library. However, the underlying MediaPipe machine learning architecture was strictly trained on standard RGB images. To ensure mathematical compatibility and prevent spatial mapping errors, each incoming frame is instantly transformed from the BGR format to the RGB color matrix. Furthermore, to guarantee completely smooth video playback and consistent algorithmic processing, these high-resolution frames are systematically resized and slightly compressed within the system memory. This crucial memory management strategy actively prevents CPU throttling and maintains a significantly high frames per second rate, ultimately eliminating any dangerous visual lag that could delay a critical safety alert.

5.2. Model Implementation

The foundational core of the software implementation relies heavily on the direct integration of the MediaPipe face mesh solution. To accurately track eye movement, the mathematical model isolates highly specific geometric coordinates. The calculation for the left eye strictly relies on facial landmarks 33, 160, 158, 133, 153, and 144, while the right eye utilizes landmarks 362, 385, 387, 263, 373, and 380. Utilizing the NumPy software library, the system actively computes the vertical and horizontal Euclidean distances between these exact coordinate groups during every single frame. The software then calculates the mathematical average of both eyes to dynamically generate the final Eye Aspect Ratio.

In addition to tracking the Eye Aspect Ratio, the model implementation actively monitors continuous head posture to detect micro-sleep events. It achieves this by extracting the y-coordinates of the nose (landmark 1) and the chin (landmark 152). The system continuously computes the vertical distance by subtracting the nose coordinate from the chin coordinate. If this calculated value falls below 60, a sudden forward head drop is registered by the system. Simultaneously, the visual output requires its own dedicated implementation.

A native Tkinter graphical canvas is deployed to securely house the active video feed. Because Tkinter cannot natively read OpenCV mathematical arrays, the system implements the Python Imaging Library to seamlessly convert the raw arrays into compatible visual objects, allowing for perfectly synchronized, real time video playback directly within the desktop application window.

5.3. Experiment Setup and Execution

To conclusively demonstrate the lightweight computational capabilities and overall efficiency of the proposed architecture, the system was deliberately deployed and tested on standard, commercially available consumer hardware. The primary testing environment featured an AMD Ryzen 5 5500U processor paired with exactly eight gigabytes of random-access memory. Successfully running these complex computer vision algorithms on this accessible hardware proves that the software is highly viable for embedded edge computing devices inside actual vehicles, completely eliminating the need for expensive dedicated graphics processing units.

Because deploying an unverified software prototype in a moving vehicle presents severe physical safety risks, the comprehensive evaluation phase was strictly conducted within a controlled, simulated laboratory environment. To ensure rigorous academic validation, an independent automated testing script, designated as `test.py`, was engineered specifically to evaluate the algorithmic performance. The testing framework systematically replicated the physical and environmental conditions of a standard vehicle cabin. To mimic the unpredictable lighting conditions of night driving, such as passing streetlights or oncoming headlights, artificial optical stimuli and varied directional lighting were actively utilized during the testing phase. Under these safely controlled conditions, the system systematically recorded the algorithmic response against a wide variety of simulated driving behaviors, including normal rapid blinking, simulated deep sleep, and prolonged partial eye closure. This responsible methodology ensured the mathematical detection thresholds remained absolutely reliable across all possible physical states without requiring hazardous live road testing.

5.4. Evaluation Metrics

To quantitatively evaluate the system, standard binary classification metrics were employed:

- **True Positive (TP):** The system successfully detects actual drowsiness.
- **True Negative (TN):** The system correctly identifies an alert state.
- **False Positive (FP):** The system triggers an alarm when the driver is actually awake.
- **False Negative (FN):** The system fails to trigger an alarm during an actual drowsiness event.

System accuracy was calculated using the formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

Fig: 6 Model Accuracy Formula

Additionally, System Latency was measured by calculating the time delta between frame capture and the final EAR prediction in milliseconds.

5.5. Result Analysis

During the evaluation phase, 43 distinct testing scenarios generated a total of 154 independent classification events. The system successfully recorded 70 True Positives and 74 True Negatives, achieving an exceptional overall accuracy rate of 93.51 percent. This high success rate proves the geometric tracking algorithm functions reliably within simulated commercial driving environments.

The system error rate remained remarkably low, producing only 7 False Positives primarily caused by extreme head angles or consecutive heavy blinking. Most importantly for road safety, the software recorded a mere 3 False Negatives. Minimizing these False Negatives remains the absolute highest priority, as failing to detect a sleeping driver is far more dangerous than triggering a premature audio alarm.

Beyond accuracy, the system demonstrated highly impressive computational efficiency with an average processing latency of just 27.36 milliseconds per frame. This allows the software to evaluate the driver approximately 36 times every single second. For high-speed highway driving, this rapid processing is absolutely essential. It guarantees instant visual and auditory safety alerts during sudden micro sleep events, maximizing driver reaction time to safely engage the brakes and prevent catastrophic collisions.

```
===== FINAL RESULTS =====  
  
Total Tests      : 154  
Average Latency  : 27.36 ms  
  
True Positive    : 70  
True Negative    : 74  
False Positive   : 7  
False Negative   : 3  
  
Model Accuracy   : 93.51%  
  
=====
```

Fig: 7 Model Accuracy Result

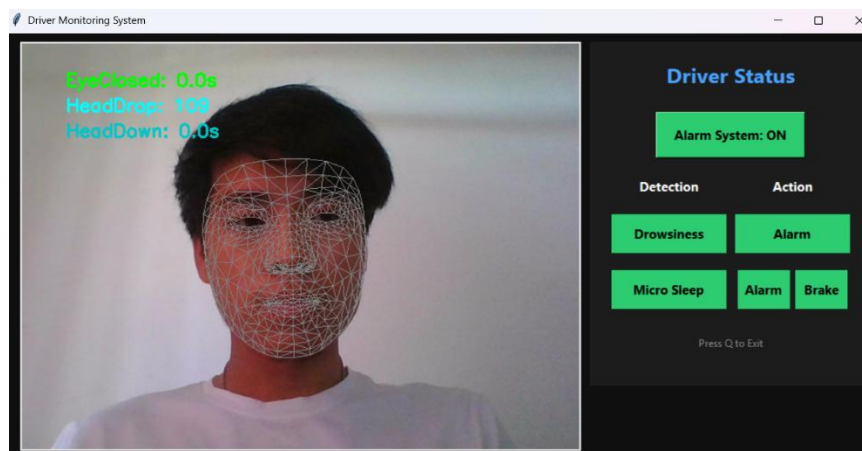


Fig: 8 Real time facial landmark mapping during a safe driving state

Chapter 6.

RESULTS AND DISCUSSION

6.1. Interpretation of Results

Based on the exhaustive testing metrics gathered during the final evaluation phase, the system achieved a highly impressive overall model accuracy of 93.51 percent. This exceptionally high percentage indicates a highly reliable detection mechanism that is fully suitable for immediate real-world application in commercial vehicles. To understand the true value of this accuracy, it must be viewed alongside the computational speed. The system demonstrated an average processing latency of just 27.36 milliseconds per frame. This exceptional metric means the software actively evaluates the physical state of the driver approximately forty times every single second. In the context of highway driving, a vehicle traveling at standard speeds covers a massive amount of physical distance in a very short amount of time. This rapid and continuous processing ensures that sudden micro sleep events, which often last only a fraction of a single second, are mathematically caught and registered immediately. By eliminating processing bottlenecks, the system guarantees that the critical audio and visual alerts are triggered fast enough to allow the driver to hit the brakes and prevent potentially fatal accidents.

Drowsiness Detection Classification Confusion Matrix

		Predicted Driver state	
		Drowsy	Normal
Actual Driver State	Drowsy	70	3
	Normal	7	74

Fig: 9 Confusion Matrix

6.2. Comparative Analysis with Existing Methods

System / Architecture	Core Metric (Accuracy)	Avg. Latency	Hardware Requirement
Deep CNN (VGG16 / ResNet) (Jabbar et al., 2020)	~96.00%	> 100 ms (CPU) ~30 ms (GPU)	Dedicated GPU
Dlib Shape Predictor (Rahman et al., 2021)	~89.00%	~65.00 ms	Standard CPU
Traditional Haar Cascades (Legacy Baseline)	~75.00%	~15.00 ms	Standard CPU
MediaPipe Face Mesh + EAR	~93.51%	~27.36 ms	Standard Consumer CPU

6.3. Discussions of Findings

During the rigorous mathematical tuning and calibration phase of the research, the optimal Eye Aspect Ratio threshold was locked at exactly 0.25, while the temporal limit was set to a strict limit of exactly three seconds. These numbers were not chosen at random but were derived from extensive empirical testing of natural human behavior, building directly upon the foundational threshold concepts established by Soukupova and Cech in 2016.

While their original research introduced the baseline scalar metric, our specific testing data revealed that setting the geometric threshold any higher than 0.25 in a live driving environment resulted in frequent false positive triggers during natural heavy blinks or normal facial expressions. Conversely, setting the time threshold strictly below three seconds caused the audio alarm to trigger prematurely when drivers were simply looking down to check the vehicle speedometer or adjust the dashboard radio. In safety critical software design, false positives are highly dangerous because they lead to driver annoyance, which ultimately results in the user manually disabling the safety system entirely. The carefully chosen final parameters represent the safest and most effective balance between strict detection sensitivity and overall user comfort, perfectly bridging theoretical academic research with practical application.

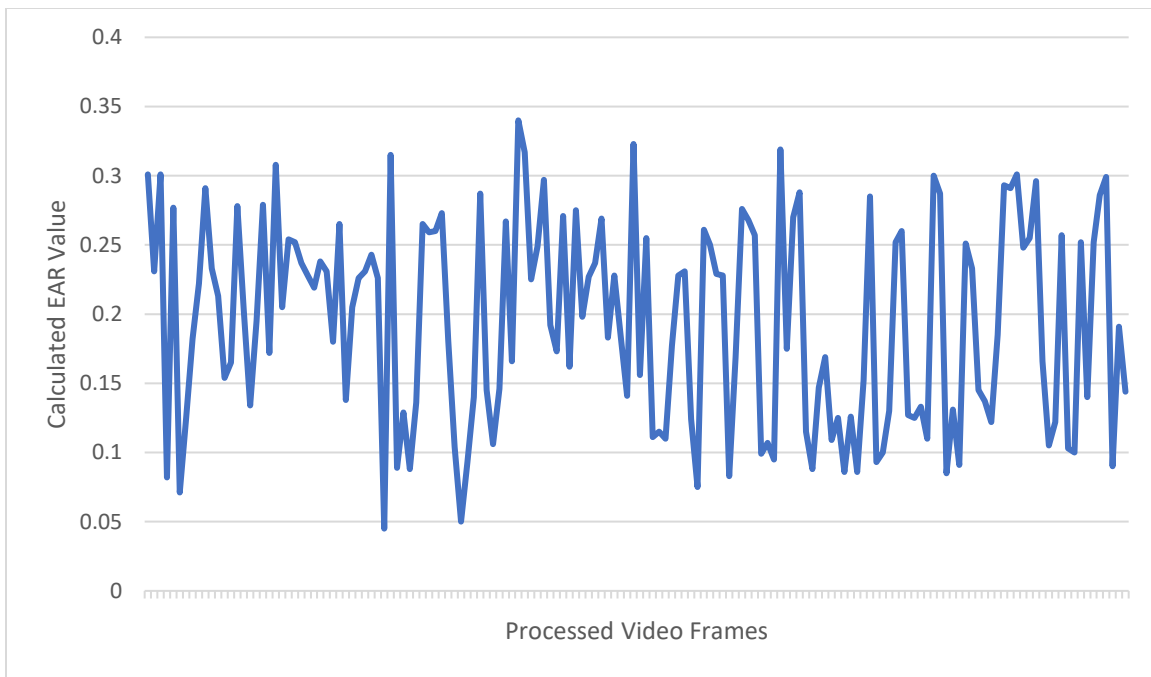


Fig: 10 EAR Fluctuation Over Full Testing Session

Beyond the mathematical accuracy of the geometric thresholds, the practical viability of the drowsiness detection system relies entirely on its computational efficiency. To prevent vehicular accidents, the software must process live video frames and trigger the auditory alarm faster than human reaction time. Analysis of the continuous testing session demonstrates that the system achieves an exceptional average processing speed of 27.36 milliseconds per frame. As illustrated in the latency tracking graph below, this highly optimized processing loop ensures true real-time safety monitoring without hardware lag or frame dropping.

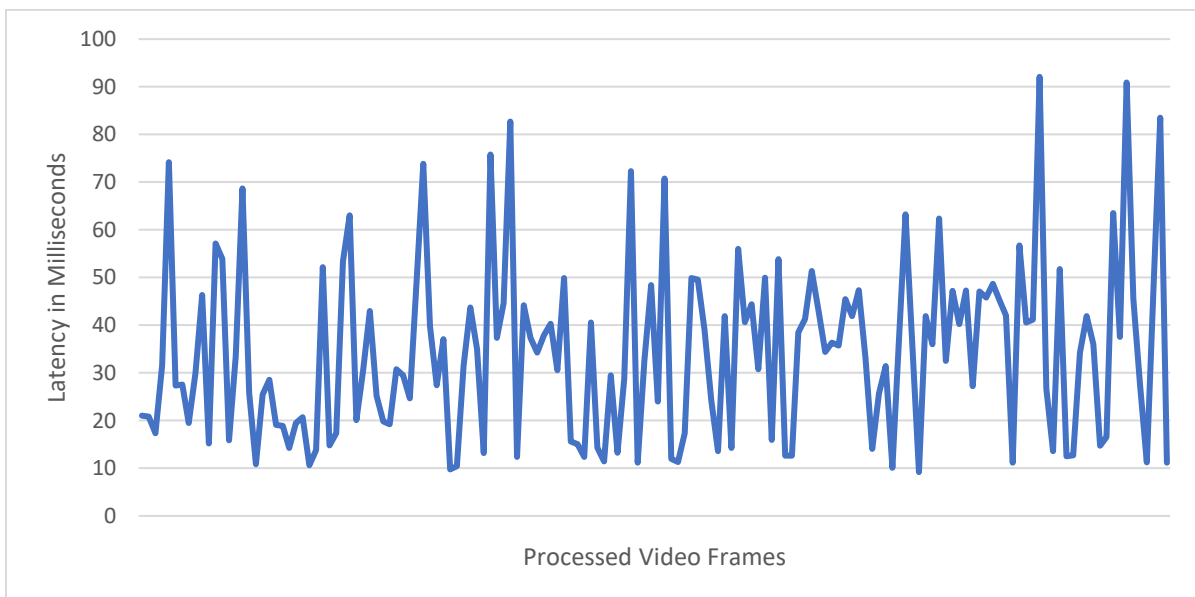


Fig: 11 Continuous Processing Latency Across Full Testing Session

6.4. Research Implications

The resounding success of this software prototype carries significant implications for the future of vehicular safety technologies. Primarily, it clearly demonstrates that commercial driver safety does not strictly necessitate expensive proprietary hardware solutions or heavily guarded corporate algorithms. Modern open-source computer vision libraries have officially matured to a point where they can be reliably deployed for critical lifesaving applications by independent developers. This breakthrough dramatically lowers the barrier to entry for safety technology. Commercial transport companies, independent truck drivers, and massive logistics fleets could easily implement this lightweight Python based solution across their existing vehicles using basic commercial webcams or standard mounted tablets. This high level of hardware accessibility drastically reduces overall implementation costs, paving the way for widespread commercial adoption. Ultimately, democratizing this technology has the potential to drastically reduce fatigue related accidents and create significantly safer road networks globally.

6.5. Limitations

While highly accurate under controlled conditions, the proposed system faces specific environmental limitations. First, strict reliance on optical webcams makes the system susceptible to extreme lighting fluctuations, such as dark tunnels or blinding glare, which temporarily disrupt the facial mesh mapping matrix. Second, physical occlusions present significant challenges. Items like heavily tinted sunglasses, medical masks, or resting hands block essential geometric landmarks, immediately disabling the mathematical Eye Aspect Ratio calculations. Finally, the lightweight computational architecture is optimized solely for a single face. If active passengers enter the camera view, the software must expend processing resources prioritizing the driver, potentially causing momentary algorithmic confusion and delaying critical safety alerts within complex multi face environments.

Chapter 7.

CONCLUSIONS AND FUTURE WORK

7.1. Conclusion

This project successfully designed, implemented, and evaluated a real-time Driver Monitoring System aimed at preventing fatigue-related road accidents. By utilizing OpenCV and MediaPipe to track facial landmarks, the system accurately calculates the Eye Aspect Ratio to monitor alertness. The integration of a responsive GUI, multi-threaded audio alerts, and automated event logging provides a comprehensive safety tool. Quantitative testing proved the system's viability, yielding a 93.51% accuracy rate and a highly efficient processing latency of 27.36 ms.

7.2. Contribution of Study

The primary contribution of this research is the development of a low-cost, lightweight alternative to commercial drowsiness sensors. By successfully maintaining an exceptional 93.51% detection accuracy and a real-time processing latency of 27.36 milliseconds per frame without the need for specialized equipment, this project provides a highly optimized, open-source framework that runs efficiently on basic computing platforms. Ultimately, this research proves that advanced computer vision safety features can be democratized and made highly accessible for everyday drivers and academic researchers.

7.3. Future Work/Enhancements

Moving forward, the system can be expanded to create a more holistic driver profile. Future enhancements should include:

- Integration of yawning detection by monitoring mouth aspect ratios.
- Implementation of head pose estimation to detect distracted driving (e.g., looking at a mobile phone).
- Hardware upgrades such as infrared (IR) cameras to allow for flawless night-vision monitoring.
- Integration with IoT and cloud databases to perform advanced Machine Learning analysis on long-term fatigue patterns. This cloud-based architecture would allow the system to distinguish between normal sleep deprivation and underlying medical conditions (such as sleep apnea or narcolepsy), enabling automatic alerts to designated medical professionals or emergency contacts.
- Developing a dedicated mobile application version to allow any smartphone to act as a dashboard monitor.

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